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An approach to the dizzy patient

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■ Although dizziness is a nearly ubiquitous complaint, few physicians other than otologists have devoted more than passing interest in recent years to the evaluation of this symptom. The attention of the otologists has been focused on the vestibular disorders about which such tests as bithermal caloric stimulation and electronystagmography (ENG) have provided much data and some valuable understanding.[1-11] Considerably less effort has been directed to the evaluation of other medical problems that may present with a similar complaint of dizziness. These include neurological, cardiac, psychiatric, hematologic, vascular, ophthalmological, and other disorders, many of which are well understood but are not as closely identified with the complaint of dizziness.

Our own experience in a university neurological outpatient clinic indicated that only a minority of patients with the complaint of dizziness suffered from vestibular disorders. Efforts to evaluate the causes of dizziness in the remaining patients were haphazard, and the identification of underlying cardiac arrhythmias, orthostatic hypotension, Bruns's apraxia of gait, and a spectrum of other disorders was sporadic and unreliable.

The need for a more systematic approach to the diagnosis of dizziness prompted the development of a model "dizziness clinic" designed to bring to bear the insights of several medical specialties, using an organized method of collecting data on each patient. The goals of the clinic, in addition to improving the

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accuracy of individual patient diagnosis, were to accumulate information regarding the frequency of the various disorders subsumed within this complaint; to learn the most useful distinguishing features (historical, physical, and laboratory), or "profile" of each of the disorders; and ultimately to derive a "least moves" strategy for future accurate diagnosis of the causes of dizziness. It was also hoped that the lessons of this "complaint-oriented" clinic approach might serve as a paradigm in other areas.

The results of our findings on the first 125 patients studied form the basis of the present report.

Methods

1] Patients. One hundred twenty-five patients admitted sequentially to the Northwestern University Dizziness Clinic comprised the study population. Patients were referred by physicians or were self-referred. Only patients who spoke English fluently were accepted; all patients had to agree before admission to return for four half-days of laboratory testing. Patients with the complaint of "dizziness" of any type were accepted without any other selection.

2] Data collection. All data were recorded on computer-oriented questionnaires in digital form. A "bypass" format was utilized to avoid the necessity of recording long strings of negative answers when a single negative answer to a master question could be used to eliminate a dependent group of questions. The questionnaire consisted of eight parts: [1] history, [2] physical and neurological examinations, [3] special dizziness examination, [4] dizziness simulation battery, [5] neuro-ophthalmological examination, [6] neuro-otological studies, [7] laboratory data, and [8] final diagnosis. Under close supervision, resident physicians in neurology completed all portions of the questionnaire except for the secretarial recording of laboratory data; all records were reviewed in toto, and final diagnoses were determined by one of us (D.A.D.). The initial examination was completed in a single morning and included a history; physical, special dizziness, and neuro-ophthalmological examinations; and dizziness simulation battery. The other parts of the questionnaire were completed later.

1] History. An accurate description of the patient's subjective experience of dizziness was obtained by first separating and classifying all complaints of dizziness into 4 types: [1] a definite rotational sensation, [2] a sensation of impending faint or loss of consciousness, [3] dysequilibrium or loss of balance without head sensation, and [4] ill-defined "lightheadedness" other than vertigo, syncope, or dysequilibrium.

Once the type of complaint had been sorted out, secondary inquiries appropriate to each type of dizziness were sought to identify related neurological, otological, cardiac, psychiatric, gastrointestinal, visual, or other symptoms. The second major area of questioning determined whether the patient's dizziness developed without any relation to position or motion or specifically when changing position or maintaining a particular posture. Finally, the intermittency or constancy of dizziness was evaluated, with particular attention paid to the occurrence of sudden circumscribed episodes or continuous symptoms. In addition, information on the overall course of the illness, medications tried, precipitating events, etc. was recorded in digital form.

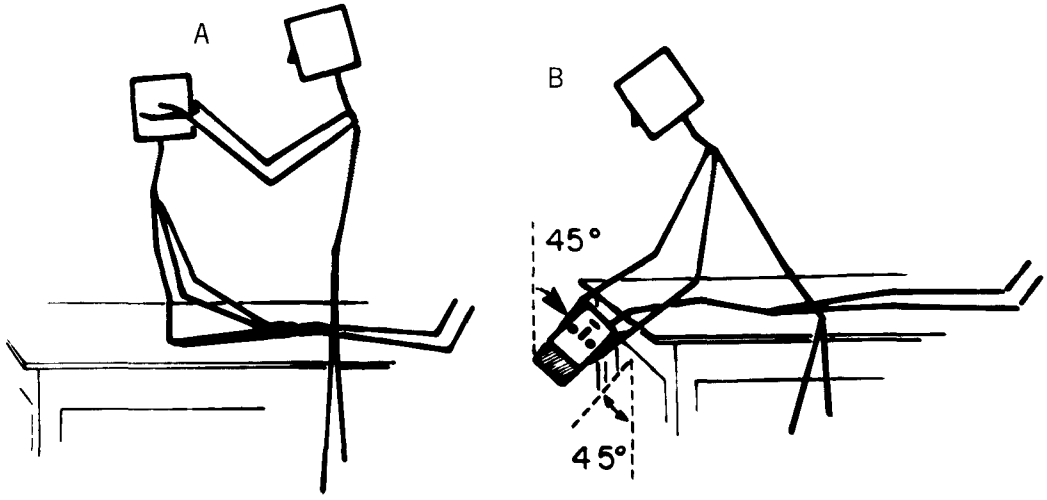
2] Physical and neurological examinations. These were performed in the standard manner. The presence of an abnormality within a system was designated on the questionnaire by a single checked box, and the details were recorded in a brief narrative. The neurological examination was subdivided so that abnormalities of mental status or cranial nerve, motor, reflex, or sensory functions could be designated in the same digital fashion and the details narratively entered.

3] Special dizziness examination. This included detailed study of heart rate and rhythm, palpation of carotid pulses, auscultation for great-vessel bruits, evaluation of grasp and suck reflexes, Quix* and past-pointing tests, Romberg's test with the head in 3 positions, and careful assessment of gait and tandem gait.

4] Dizziness simulation battery. This group of tests evolved as one of the most important components of the entire study. It was designed to expose the patient to a number of situations

* The patient stands, extends both arms, points with his index fingers at the examiner's index fingers (held at shoulder height), and then closes his eyes. Drift of both hands consistently in either direction is abnormal.

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The Nylen-Bárány maneuver for positional vertigo and nystagmus. The patient is moved abruptly from a seated [A] to a prone [B] position, with his head hanging 45° below the horizontal and rotated 45° to one side. He is observed for the development of nystagmus and vertigo.

that commonly trigger various types of dizziness in susceptible individuals and to reproduce for every patient each of the 4 types of dizziness. This enabled the physicians to communicate more precisely than with words what was meant by each type of dizziness; it enhanced the ability of the patient to identify the sensation most closely simulating his own complaint. Nine maneuvers were carried out; following each, the patient was asked if he was dizzy and, if so, whether this sensation of dizziness matched his own complaint. The maneuvers included: blood pressure determinations, lying and standing (immediately and after three minutes); a standard Valsalva maneuver (forced expiration at 40 mm. Hg for fifteen seconds); carotid sinus stimulation for ten seconds; head turn, sitting and standing with eyes open and closed; sudden turn when walking; hyperventilation (HV) for three minutes; a Nylen-Bárány test* for positional vertigo and nystagmus; and a standard Bárány rotation (10 rotations within twenty seconds). The patient was also asked to perform any other maneuvers that especially triggered his own dizziness. When dizziness was produced by any of the maneuvers, stabilizing procedures, including lying, sitting, touching the examiner's finger, etc., were evaluated for effectiveness in relieving symptoms.

5] Neuro-ophthalmological examination. This included routine tests of visual and ocular motor function, as well as a special test of vestibulo-ocular and pursuit movements.[12] Where appropriate, ophthalmodynamometry was performed.

6] Neuro-otological studies. The examination of the vestibular system was carried out with electronystagmographic recording of eye movements, using the methodology recommended by the ENG Study Group Committee.[13] Beckman Biopotential miniature electrodes were affixed lateral to the outer canthi of the eyes, and a grounding electrode was placed on the forehead. Recordings were made on a Beckman Type R Dynagraph with 4 channels equipped for direct-current rectilinear recording of corneoretinal potentials and the first derivative of this function; this provided tracings of both the actual eye positions and the speed of ocular movements. Ocular motility was first tested by a sequence of maneuvers including opticokinetic nystagmus tests. Next, spontaneous nystagmus and positional nystagmus in response to the Nylen-Bárány maneuver were sought.[11] Then, the bithermal caloric test,

* The Nylen-Bárány maneuver for positional vertigo is performed by seating the patient on the edge of an examining table and abruptly having him lie down with the head hanging 45° backward and turned 45° to one side. The patient is observed for the development of vertigo and nystagmus and the time of onset, duration, and direction of the quick phase of the nystagmus. The test is then repeated with the head turned to the opposite side (see illustration).

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TABLE 1

DIZZINESS CLINIC LABORATORY TESTS

Hematology, blood chemistry, urinalysis

Complete blood count and sedimentation rate
FBS, BUN, total protein, albumin-globulin ratio, T_4-by-column, and serology
Urinalysis

Radiologic studies

Skull series with Stervers' views
Cervical spine, 6 views
Chest survey

Electrodiagnostic tests

Electroencephalogram
Nerve conduction velocities (lower extremities)
Electrocardiogram with rhythm strip

Psychometric testing

MMPI

Optional tests

Polycyclic tomograms of petrous bones
Cardiac monitoring
Five-hour glucose tolerance test
Complete battery of psychometric tests

using a modification of the Hallpike technique,[13] was carried out. Finally, the patient was requested to hyperventilate for three minutes and another rapid positional test was performed.

An extensive battery of hearing tests was given to all patients regardless of the presence or absence of auditory complaints. These included air and bone conduction, pure-tone threshold, speech reception threshold (SRT), speech discrimination, threshold tone decay (adaptation), Bekesy audiometry, and short-increment sensitivity index (SISI) tests.

Where applicable, alternate binaural loudness balancing (ABLB), simultaneous binaural median-plane localization (SBMPL), and other tests of hearing for filtered, masked, and binaural speech were performed.[14,15]

7] Laboratory tests. The battery of routine studies was divided as follows: hematologic, blood chemistry, and urine studies; radiologic studies; electrodiagnostic tests; and psychometric evaluation (Table 1). In addition, when appropriate, a glucose tolerance test, a complete battery of psychometric studies, polycylic tomograms of the petrous bones, or any other studies deemed necessary were performed. This occasionally required hospitalization of the patient.

8] Final diagnosis. Review of all findings and laboratory results was carried out when the patient returned to the clinic. Diagnoses were classified as: "major" when the symptoms resulting from the disorder accounted fully for the patient's complaint of dizziness, "contributing" when the disorder either increased the disability or was an underlying cause of the specific manifestations producing dizziness (e.g., diabetes underlying neuropathy), or "unrelated" if the abnormality, however serious, had no relation to the patient's dizziness. In many cases the clear causal relationship between the patient's complaint and the abnormal clinical or laboratory findings made the diagnosis self-evident; examples were syncopal dizziness associated with marked orthostatic hypotension or rotational vertigo associated with a severe labyrinthine disorder. In other instances diagnoses were based on patterns, or profiles, of both normal and abnormal findings as seen, for example, in patients with multiple sensory deficits or the hyperventilation syndrome. A discussion of the hierarchical ordering of criteria for each diagnosis is beyond the scope of this paper and will be presented elsewhere; some examples are given below.

Results

One hundred twenty-five patients sequentially admitted to the clinic over almost a two-year period comprised the study population. Of these, 21 were rejected because of incomplete data, leaving 104 for whom sufficient data were available to attempt formulation of a diagnosis (Table 2). Secure diagnoses could be made in

TABLE 2

DIZZY PATIENTS – POPULATION STUDIED

Total number of patients	125	
Rejected (inadequate data)	21	
Accepted for study	104	
Patients with 1 major diagnosis	81	91%
Patients with 2 major diagnoses	12	
Inappropriate referrals	2	
Uncertain diagnosis	9	9%

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TABLE 3

CAUSES OF DIZZINESS: MAJOR ETIOLOGIES*

	Percent
Peripheral vestibular disorders	38
Hyperventilation syndrome	23
Multiple sensory deficits	13
Psychiatric disorders	9
Uncertain diagnosis	9
Brainstem cerebrovascular accident	5
Neurological disorders, other	4
Cardiovascular disorders	4
Other	
Multiple sclerosis	2
Visual disorders	2
Endocrine disorders	1
Excessive awareness of normal sensation	1
Cervicogenic disorder	0
Seizures	0

* Total is greater than 100% because 12 patients had 2 diagnoses.

95 patients, or 91%. Of these, 81 patients had 1 major diagnosis, 12 had 2 major diagnoses, and 2 were regarded as inappropriate referrals who, although referred for dizziness evaluation, denied the complaint of dizziness. No final diagnosis could be arrived at in 9 patients, or 9% of the study population. In this group, some patients had no relevant abnormalities while others had too many, none of which could be clearly identified as the cause of the presenting complaints.

In Table 3 the relative frequencies of occurrence of 13 major etiologies of dizziness are tabulated. The occurrence of 2 major etiologies in 12 patients accounts for the total of 112%. Vestibular disorders accounted for the largest proportion of complaints of dizziness although, as will be shown below, this in itself is a heterogeneous group. The hyperventilation syndrome and the syndrome of spatial disorientation due to multiple sensory deficits together were responsible for almost as many cases of dizziness. It is worth noting that cerebrovascular disease involving the brainstem, either with fixed infarctive lesions or with transient ischemic attacks, was an uncommon cause of dizziness in this study.

Peripheral vestibular disorders. Thirty-nine patients had significant peripheral vestibular disorders as a major cause of dizziness; of these, 32 had vestibulogenic dizziness alone while 7 had an additional cause of dizziness (Table 4). The hyperventilation syndrome was responsible for the second type of dizziness in 5 of these patients. They complained of light-headedness as well as vertigo.

The diagnosis of a peripheral vestibular disorder was, typically, applied to a patient who complained of unmistakable rotational vertigo, frequently with nausea and at times with vomiting. The absence of evidence of neighboring brainstem deficits was a prerequisite for the diagnosis of a peripheral vestibular disorder.[16-18] Bárány rotation and caloric testing often reproduced the patient's symptoms exactly, as did the Nylen-Bárány positional test in those with positional vertigo. In many cases, the Quix, past-pointing, and Romberg tests revealed the patient's predilection to turn or tilt in one direction. The caloric tests often demonstrated appropriate peripheral vestibular abnormalities;[5,8,9,11,19,20] in others, electronystagmographic recordings documented the development of positional nystagmus or the occurrence of spontaneous nystagmus.[21-23] Audiometric findings aided in distinguishing peripheral labyrinthine disorders from brainstem involvement, particularly when the cochlea was affected along with the labyrinth.[14]

An outstanding finding was the fact that positional vertigo occurred twice as frequently as any other vestibular disorder. To avoid con-

TABLE 4

PERIPHERAL VESTIBULAR DISORDERS

	No. of patients
Benign positional vertigo	12
Acute and recurrent peripheral vestibulopathy	6
Acute peripheral vestibulopathy (1 attack)	4
Ménière's disorder	4
Chronic labyrinthine imbalance	3
Other	12
Total	41 (38%)*

* Total number of patients was actually 39 but 2 had 2 types of vestibular disorders.

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fusion, positional vertigo was defined as a true rotational sensation occurring only on change of position and for a brief duration (minutes). Cases in which spontaneous vertigo occurred at rest but was made worse by change of position or in which positional symptoms developed some time after a more prolonged bout of spontaneous vertigo had subsided were not included in this group. All of our cases had "benign" postural vertigo, as described by Citron and Hallpike.[21] Classically, the nystagmus and vertigo in this form occur after a latent period of a few seconds following assumption of the provocative posture and adapt after a minute or so or fatigue upon repetition of the maneuver.[21-23] In the "malignant" variety, seen with space-occupying, demyelinating, or other lesions of the posterior fossa, the onset of nystagmus is immediate, adaptation or fatigue of nystagmus does not occur, and subjective vertigo is minimal or absent.[24]

"Acute peripheral vestibulopathy" was defined as a single bout of spontaneous vertigo lasting for hours or days with a variable residual deficit of one peripheral vestibular system (labyrinth or nerve or both). "Acute and recurrent peripheral vestibulopathy" was similar but recurred at least once after the initial episode.

Ménière's disorder was diagnosed if acute and recurrent peripheral vestibulopathy was accompanied by hearing loss and tinnitus, often with a characteristic "cochlear deficit" pattern in the audiometric studies.[25-27] The small proportion of patients seen with this disorder in our study is of interest and contrasts strikingly with the reputation enjoyed by this condition as a leading cause of dizziness and with the extensive literature regarding its diagnosis and treatment.[25-29] It is worth noting parenthetically that many patients ultimately diagnosed as having other conditions had initially entered the study with a diagnosis of Ménière's disorder.

Chronic vestibular imbalance was diagnosed when a patient with momentary vertigo or disorientation on sudden change of head position was shown to have significantly asymmetric caloric responses on bithermal testing of the two ears and no other abnormality.

The "vestibular-other, undetermined" group included patients with clear symptoms of vertigo in whom bilateral hypoactivity of the labyrinths occurred without explanation or in whom a central (brainstem) vestibular disorder could not be reliably distinguished from a peripheral one because of the paucity of residual neurological findings at the time of examination. Cerebellopontine angle tumors, included in the classification of peripheral vestibular lesions, were not found in this series.

Hyperventilation syndrome. The hyperventilation (HV) syndrome was diagnosed in 24 patients, or 23% of the study series (Table 5). Fifteen of the 24 had significant psychiatric disorders but were included in this group because the complaint of dizziness was directly attributable to the symptoms of HV. Five patients, as previously noted, had additional significant vestibular disorders; 4 had HV unassociated with any other major etiology of dizziness.

The majority of these patients—13—had acute attacks of HV producing relatively circumscribed episodes of light-headedness, frequently with circumoral and digital paresthesias.[30-33] A smaller number—10—had longer lasting symptoms of light-headedness of more gradual onset and offset; the symptoms in both groups were precisely reproduced by HV but none of the other maneuvers in the dizziness simulation battery. Two patients manifested a syndrome, not previously described, of positional vertigo occurring only after HV. In these cases the Nylen-Bárány maneuver produced vertigo and nystagmus only after they had hyperventilated in the standard fashion; the patient's complaints were, by history, of positional vertigo associated with HV symptoms. We have confirmed this phenomenon in

TABLE 5

HYPERVENTILATION SYNDROME

	No. of patients
Acute attacks of HV	13
Chronic dizziness, reproduced by HV	10
HV producing positional vertigo	2
Total (15 had associated psychiatric disorder)	25 (23%)

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TABLE 6

MULTIPLE SENSORY DEFICITS

Causes of deficits	*No. of patients*
Peripheral neuropathy	12
Cervical spondylosis	10
Vestibular abnormality	9
Visual impairment	5
Orthopedic problem (motor)	1
No. of combined deficits	
Two sensory deficits	7
Three sensory deficits	5
Four sensory deficits	2
Total	14 (13%)

several other patients not included in this series and now include HV followed by positional testing in our routine ENG battery. The mechanism of this interesting phenomenon has not yet been ascertained but is under study in our laboratory.

Multiple sensory deficits. Fourteen patients (13%) were found to have multiple sensory deficits producing dizziness. Four of these patients had an additional major cause of dizziness, as noted in Table 6.

The abnormalities comprising this syndrome included two or more of the following: visual impairment (not correctible), neuropathy, vestibular deficits, cervical spondylosis, and orthopedic disorders interfering with ambulation. Typically, patients manifesting this syndrome were elderly, often diabetic. They noted no dizziness when lying, sitting, or standing still but complained of light-headedness (type 4) when walking and particularly when executing a turn. They were not ataxic but felt transiently confused when walking; they rarely, if ever, complained of falling. Dizziness was most closely reproduced by walking or walking and turning quickly and, in some, by turning the head when standing. Touching the examiner's finger lightly provided sufficient additional sensory information to relieve symptoms markedly in most patients. The appropriate clinical tests revealed evidence of peripheral neuropathy accompanied by slowed nerve conduction velocities in the lower extremities. Visual loss was often due to cataracts, and vestibular abnormalities consisted of inadequate responses or imbalance of the peripheral labyrinths rather than subjective spontaneous vertigo. The significance of cervical spondylosis was sometimes difficult to judge but was considered to be related when increase of symptoms on head turning was associated with marked degenerative changes seen on roentgenograms of the cervical spine.

Of 14 patients with this diagnosis, 12 had peripheral neuropathy; 10, cervical spondylosis; 9, vestibular impairment; 5, visual loss; and 1, a contributing orthopedic disturbance. All patients in this group were elderly, and our experience both in this series and in patients seen subsequently suggests that multiple sensory deficits may be the most important cause of vaguely described dizziness in an aged population. The observation that multiple sensory deficits may result in the subjective complaint of dizziness is little appreciated; the mechanism by which these deficits produce dizziness is discussed below.

Psychogenic dizziness. Dizziness caused by a psychiatric disturbance was diagnosed in 9 patients, or 9% of the study series (Table 7). Six of these patients were primarily depressed and anxious and characteristically complained of vague light-headedness (type 4). Dizziness had often been present for five or ten years and was usually continuous rather than episodic. Most patients admitted to the symptoms of anxiety–panic, fright, feeling of remoteness, and perspiration–and of depression–difficulty concentrating; loss of energy, interest, and

TABLE 7

PSYCHOGENIC DIZZINESS

	No. of patients
Neurotic–anxiety, depression, hysteria	4
Psychotic–bizarre complaint	4
Astatic–abasia	1
Total (In addition, 15 had hyperventilation and psychogenic disturbance)	9 (9%)

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appetite; fatigue; and depression—suggested in the questionnaire. The dizziness simulation battery demonstrated that either most of the maneuvers or none at all reproduced the patient's symptoms. The examining physician's clinical impression of anxiety and depression was usually supported by the Minnesota Multiphasic Personality Inventory (MMPI), which showed an abnormal elevation of scores in the "neurotic triad" of depression, hysteria, and hypochondriasis.

Three patients were psychotic, 2 with chronic schizophrenia requiring institutional care in the past and 1 with a probable diagnosis of schizophrenia. Symptoms of dizziness were variable and numerous and did not fit any of the recognizable patterns, an observation consistent with other evidence of a thought disorder.

It is important to emphasize that in none of these patients were the symptoms of hyperventilation exclusively or primarily responsible for the complaints of dizziness. Rather, the dizziness was equated with "loss of energy," "difficulty thinking," "mental fuzziness," and similar symptoms. There were, as previously noted, 15 patients with the hyperventilation syndrome related to underlying psychiatric disturbances. These patients are not included in the psychogenic dizziness group, however, since the mechanism of production of the dizziness, and the type of complaints, differed.

In addition to those patients in whom psychiatric disorders were primarily responsible for the complaints of dizziness, others had, of course, developed secondary depressive reactions, symptoms of anxiety, and somatic preoccupations as a result of the long-standing poorly understood disability produced by their dizziness. Still others had a variety of psychiatric disturbances that were neither the cause nor the result of their dizziness but an unrelated incidental occurrence.

Cerebrovascular accidents. Five patients (5%) were found to have cerebrovascular accidents producing dizziness. Four were over 60 years of age and 1 was 48. In each case the onset of dizziness was abrupt; all complained of true rotational vertigo (type 1). One patient had had a massive brainstem infarction with severe cranial nerve, cerebellar, and long tract signs; in the others evidence of additional neurological involvement was limited to diplopia, facial weakness and numbness, unilateral hyperreflexia or weakness of the extremities, or mild impairment of cerebellar function. Two patients had diabetes; 1 had coronary insufficiency.

The small percentage of patients with cerebrovascular accidents probably reflects accurately the infrequency with which a primary complaint of dizziness is due to cerebrovascular disease. This is not to suggest that basilar-vertebral insufficiency or brainstem infarctions do not produce dizziness, since it is well known that vertigo due to ischemia of the vestibular nuclei often occurs in these conditions. It is important to stress, however, that vertigo due to ischemia rarely occurs as an isolated phenomenon in the absence of neurological deficits resulting from damage to adjacent brainstem structures.[16-18] There is also no evidence that either cerebrovascular disease of the carotid circulation or "generalized cerebral arteriosclerosis" produces the complaint of dizziness by some other (nonvestibular) mechanisms.

Cardiovascular disorders. Four patients (4%) had dizziness due to impairment of total blood flow to the brain. Two patients had orthostatic hypotension and hypersensitivity of the carotid sinus, 1 with varying tachyarrhythmias as well. One patient had micturition syncope, while the fourth had marked anemia. All of these disorders except anemia produced sudden episodes of syncope-like sensations (type 2), sometimes quite brief but at other times resulting in actual loss of consciousness.[34-37] The demonstration of the blood pressure or pulse rate changes under appropriate circumstances, accompanied by symptoms matching the patients' complaints, established the diagnosis in each case.

Multiple sclerosis. Two patients (2%) with dizziness were found to have multiple sclerosis, both with complaints of true rotational vertigo (type 1). Although only a small percentage of dizzy patients are likely to have multiple sclerosis, a very large proportion of patients with multiple sclerosis experience dizziness of one form or another at some time during their illness. This may be vertigo on the basis of demyelination in the region of the vestibular nuclei or connections,[12] light-headedness due to multiple sensory deficits, gait impairment

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secondary to ataxia, a visual problem due to interruption of vestibulo-ocular connections, diplopia, or dizziness due to any of a number of other mechanisms. A further study of dizziness in multiple sclerosis patients is under way in our laboratory.

Neurological disturbances, other. Four patients with other neurological disorders leading to dizziness, or 4%, were seen in the present series. Two had Bruns's apraxia of gait with impairment of ambulation on the basis of frontal lobe disease, most likely degenerative in origin (Alzheimer's disease). One patient had early parkinsonism and interpreted the motor impairment as dizziness. These 3 patients complained of dysequilibrium with difficulty in controlling the legs and balancing (type 3). The fourth had suffered from vague light-headedness ever since a partial temporal lobectomy and the clipping of a ruptured cerebral aneurysm; she also had multiple sensory deficits, including impairment of sound localization, accounting for her symptoms.

Visual disorders. Two patients had visual disturbances producing dizziness. One showed the clinical picture of difficulty with ambulation following successful cataract extraction. In this patient severe cervical spondylosis and marked high-tone hearing loss with impairment of sound localization were additional factors in her dizziness. The distortions of vision due to aberrations produced by strong positive lenses ("pincushion effect," "Jack-in-the-box phenomenon," ring scotoma, and environmental motion opposite to head motion)[38] produce disorientation that is poorly tolerated in patients with any deficit of sensory modalities besides vision. We have seen and studied other patients with this disorder who are not included in this series; efforts to evaluate the critical factors producing this syndrome, and to identify them preoperatively, are being undertaken.

The other patient had diplopia secondary to long-standing ocular muscle imbalance with vague light-headedness upon gaze in directions that emphasized her diplopia.

Other diagnoses. One patient was found to have "excessive awareness of normal sensations," complaining of momentary dizziness on rapid change of position but with objective evidence of dizziness that did not exceed normal experience. One patient with myxedema, clinically evident and confirmed by a T_4-by-column test, was tentatively thought to have some relation between her hypothyroidism and symptoms of dizziness,[39] although psychic factors also played a significant role.

Comment

The most striking observation of this study was the unexpectedly high proportion of patients in whom a reasonably secure diagnosis could be reached: 91% of those who completed the battery of tests. In the majority of these patients conventional diagnoses were arrived at by the persistent, systematic application of well-established diagnostic techniques in an organized sequence. For example, orthostatic hypotension and hypersensitivity of the carotid sinus are well-known causes of dizziness but were recognized as the specific causes of dizziness in particular patients only when they were specifically sought.

In an additional large number of patients, however, certain special testing sequences assembled for this study were instrumental in arriving at a diagnosis. The dizziness simulation battery proved to be of great value in identifying accurately the type of dizziness experienced by many patients when verbal descriptions did not suffice. The "vestibulo-ocular, pursuit (V-O, P)" test, which differentiated the impairment in several of the systems for the control of ocular motility, proved of value,[12] as did several modifications of the electronystagmographic test battery.

Diagnostically, disorders of the vestibular system proved to be the most common causes of dizziness, accounting for 38% of all patients. The most common vestibular disorder was benign positional vertigo, accounting for three times as many complaints of dizziness as Ménière's disorder. The absence of cases of acoustic neuroma is also worth noting, suggesting that it is a rare, albeit important, cause of dizziness. Of the other vestibular disorders, we would call attention to use of the term "peripheral vestibulopathy" to describe acute or recurrent episodes of vertigo due to extramedullary disorders of the vestibular system. However dissonant the term may be, it has been used in preference to the more familiar "vestib-

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ular neuritis," "vestibular neuronitis," "acute labyrinthitis," "epidemic vertigo," "viral labyrinthitis," and other terms that suggest a knowledge of the exact site or nature of the pathological process where both remain uncertain.[40-43]

The importance of the hyperventilation syndrome as a cause of dizziness requires some emphasis since this entity accounts for almost one-fourth of all the patients in our series. Although the underlying cause of hyperventilation is most frequently psychiatric, it is the hyperventilation itself that produces the patient's symptoms of dizziness. Since a recognizable physiologic mechanism likely underlies the symptoms, i.e., vasoconstriction of cerebral vasculature,[31] there is a realistic possibility of utilizing therapeutic methods that deal with the HV rather than primarily attacking the underlying psychiatric disturbances.

Finally, the major significance of multiple sensory deficits involving vision, proprioception, touch and pressure sensation, and vestibular function in the causation of clinical dizziness should be stressed. Although it is obvious that deprived of accurate sensory information any organism would become disoriented, it is not widely enough appreciated that in the human being this often produces a sensation referred to as dizziness by its sufferers.

Discussion

In the present study we have accepted the basic premise that patients complaining of dizziness make this complaint primarily when they are uncertain of their position or motion in relation to the environment. Such uncertainties may be the result of perceptual, integrative, or motor (ambulatory) disorders. Spatial disorientation has been well studied in many aerospace investigations[44-47] where subjects with normal sensory mechanisms have become dizzy when subjected to "unphysiologic" maneuvers resulting in uninterpretable discrepancies between various sensory modalities.

Unlike the aerospace investigations, we have been concerned with spatial disorientation occurring under physiologic environmental conditions in individuals suffering from abnormalities of the orienting mechanisms.

By accepting the broadest concept of dizziness as a disorder of human spatial orientation, the present study has recognized the unnamed causes of dizziness where perceptual impairment plays a role, as well as specific diagnostic entities such as Stokes-Adams attacks or hypoglycemia. A second feature of this series was the systematic application of a comprehensive battery of diagnostic studies to all patients complaining of dizziness. In many other investigations individual causes of dizziness, most notably those related to the vestibular system, were well studied. In some of these studies, all patients complaining of dizziness were evaluated from the viewpoint of vestibular disease,[5] but in most only those with true rotational vertigo were even considered.[48-52] Elsewhere, hyperventilation, orthostatic hypotension, cervical spondylosis, peripheral neuropathy, etc. have been studied in far greater detail than in the present study but only as isolated disease entities rather than in terms of their contributions to the overall problem of dizziness.

We are unaware of other studies in which all of the recognizable causes of dizziness considered here were assembled in an organized investigation. For this reason it is difficult to compare the results of the present series with estimates of the frequencies of various causes of dizziness reported elsewhere in the literature. Spector has commented on the disparity between studies of dizziness of different authors;[53] a survey of the recent literature confirms the impression that each report deals with a limited segment of the dizzy population[49,50,52] rather than encompassing the entire problem.

It would be misleading to imply that this investigation has resulted in an understanding of the cause of Ménière's disease or in a "cure" for dizziness. Rather, it has provided a working classification of the many causes of dizziness and a means for arriving at the appropriate diagnosis in each case. It has provided some perspective of the relative importance of certain types of dizziness, e.g., hyperventilation, and of the relative rarity of others, e.g., Ménière's disorder.

This study should serve as a point of departure for further investigation of clearly isolated groups of patients with discrete disorders. For some of the groups treatments are already well established, e.g., the cardiac

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arrhythmias; for others, meaningful studies of treatment can only be effectively begun when the groups are cleanly separated. In addition, we have obtained the base-line information needed to attempt to develop more efficient and less time-consuming means of establishing diagnoses in dizzy patients. It is already evident to us that there is considerable redundancy in the diagnostic schema that has been utilized in this study, and an effort is under way to develop a system of branching logic that will identify appropriate diagnostic entities using a least moves strategy.

It is necesary to identify the weaknesses, as well as the strengths, of a study such as this. Resident physicians in neurology, under supervision, performed most of the examinations; training them in the concepts and approaches of the evaluation of the dizzy patient took many sessions, although all were moderately sophisticated in medicine and neurology before beginning this aspect of their training. Several of the patients found the number of appointments too inconvenient and failed to return for the entire battery of tests. Finally, the validity of certain of the assumptions used in arriving at diagnoses can be challenged. For example, we have accepted the concept that patients with some combination of peripheral neuropathy, visual impairment, vestibular disorders, and cervical spondylosis who complain of dizziness on walking and turning but not when resting have their dizziness on the basis of multiple sensory deficits. It is by no means clear how much of each of these deficits is needed to produce the sensation of dizziness, the variation in tolerance from individual to individual, or the extent of similar deficits in a nondizzy age-matched population. Such evaluations remain for the future.

Summary

To apply a broad medical approach to the problem of the dizzy patient, a model dizziness clinic was set up at Northwestern University Medical School. All patients with a complaint of dizziness were accepted, and data pertaining to relevant medical, neurological, otological, psychiatric, cardiac, and other problems were collected in digital form according to a highly organized protocol. Special features of the study included the separation of patients' complaints into four types and the use of a dizziness simulation battery of maneuvers to reproduce each type of dizziness for identification by the patient.

The first 125 patients admitted to the clinic form the basis of this report. Sufficient data to attempt to formulate a diagnosis were collected on 104 patients; secure diagnoses could be reached in 91%. Vestibular disorders were found in 38%; positional vertigo was the most common in this group, while Ménière's disorder comprised only 5% of the total. The hyperventilation syndrome was responsible for 23% of the complaints of dizziness and multiple sensory disorders were responsible for 13%. Primary psychiatric disorders, cardiovascular and cerebrovascular disease, multiple sclerosis, and other miscellaneous disorders accounted for the remaining diagnoses.

Our experience suggests that a highly organized, complaint-oriented clinic may be far more efficient in diagnosing the causes of a complex problem such as dizziness than a less-ordered physician-patient transaction, although some of the testing is superfluous or redundant.

Through the insights of this experience, dizziness is seen in its broadest sense as a disorder of human spatial orientation rather than as any particular disease entity. Although vestibular disease is a common cause of the complaint of dizziness, to equate dizziness with vestibular disease is contrary to the understanding of many patients and would eliminate more than one-half of the actual causes of this complaint from consideration.

Modification of the strategy for the diagnosis of dizziness is under development to minimize the number of steps required to arrive at a diagnosis and to eliminate procedures that may be unnecessary in particular cases without the loss of diagnostic accuracy.

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